

SMART INDIA HACKATHON 2026



- **Problem Statement ID :- 26101**
- **Problem Statement Title:- Develop an AI enabled learning platform that identifies competency gaps recommends personalized training through integration with the iGOT Karmayogi ecosystem capable of generating Quizzes Multiple choice questions (MCQs) from uploaded learning materials to strengthen capacity building in India's Official Statistical System.**
- **Theme:- Smart Education**
- **PS Category:- Software**
- **Team ID:- 71**
- **Team Name :- Fusion Nexus**

Develop an AI enabled learning platform that identifies competency gaps recommends personalized training through integration with the iGOT Karmayogi ecosystem capable of generating Quizzes Multiple choice questions (MCQs) from uploaded learning materials to strengthen capacity building in India's Official Statistical System.



❑ Proposed Solution :

An AI-powered Competency Intelligence Platform for India's Official Statistical System that replaces generic training with role-specific, competency-driven learning.

Core Components

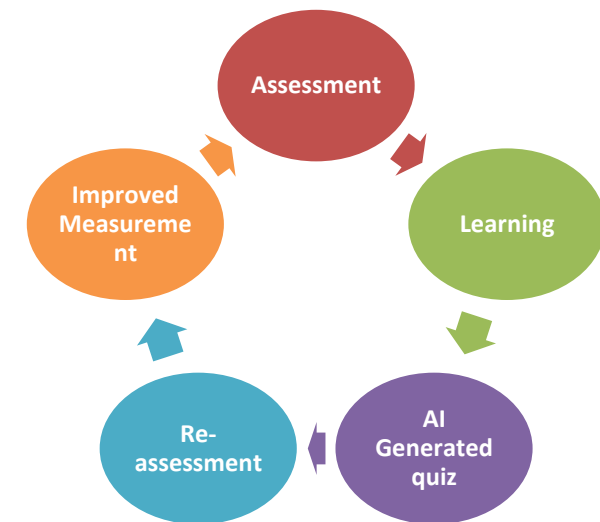
- **Adaptive Gap Engine:** Maps job roles to required skills and uses adaptive testing to pinpoint precise competency gaps.
- **iGOT Karmayogi Linkage:** Automatically recommends targeted courses from iGOT Karmayogi and Capacity Building Programmes (CBPs) based on identified skill gap.
- **RAG-Driven Quiz Generator:** Uses Retrieval-Augmented Generation(RAG) to generate grounded, hallucination-free MCQs directly from uploaded statistical PDFs and manuals.
- **Workforce Analytics:** Tracks real-time competency levels, learning progress, and skill gaps across all organizational roles of India's Official Statistical System.

Value Proposition

- **Targeted:** Shifts focus from generic course completion to measurable skill acquisition.
- **Automated:** Converts official documents into custom assessments instantly with verified source citations.
- **Closed-Loop:** Delivers an Assess → Learn → Reassess → Measure growth lifecycle.

❑ Innovation & Uniqueness

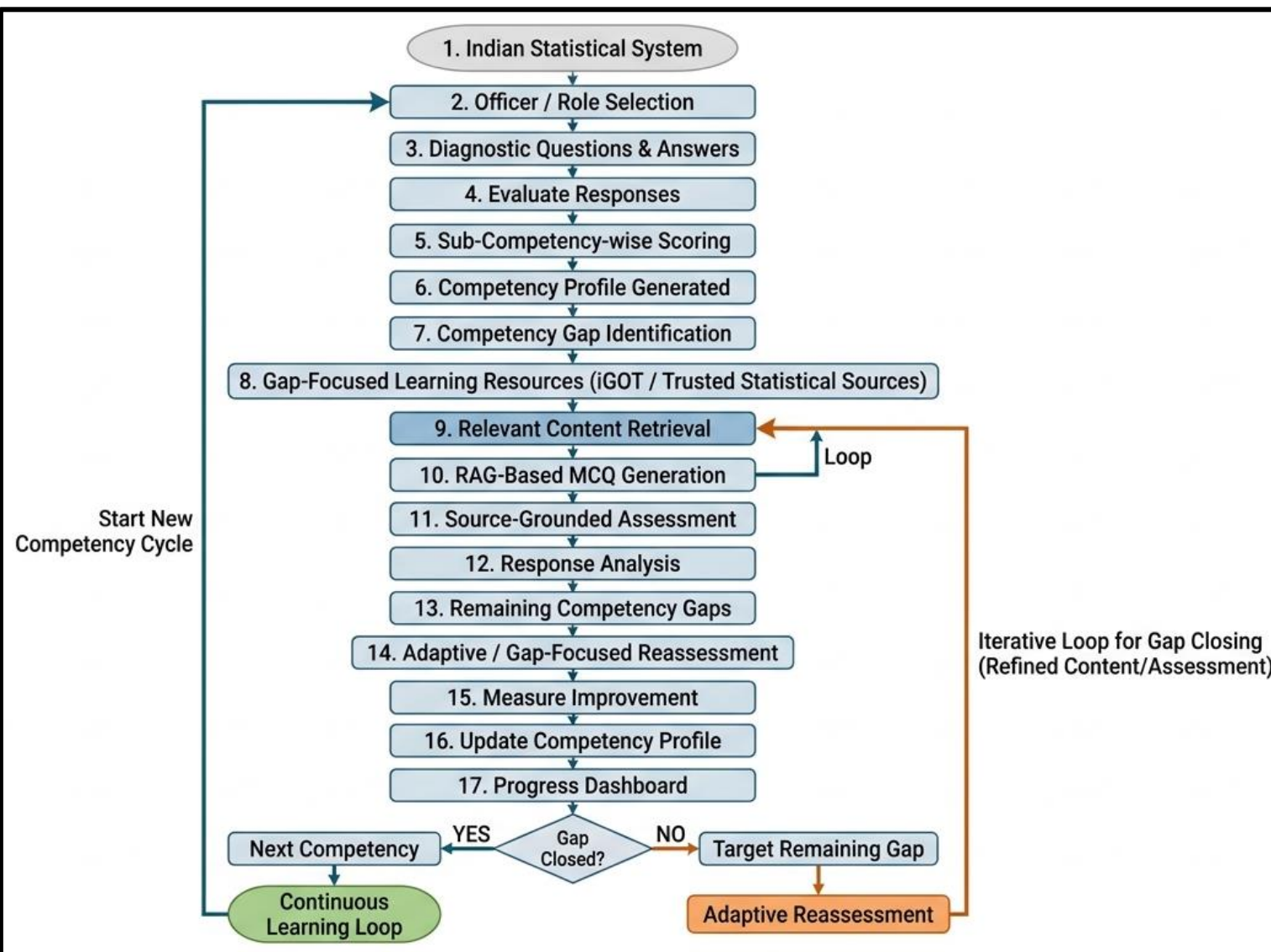
- ❖ RAG-Based MCQ Generation.
- ❖ Source Grounded Assessment.
- ❖ Adaptive Reassessment.
- ❖ Gap Focused Questioning.
- ❖ Closed Loop Learning.

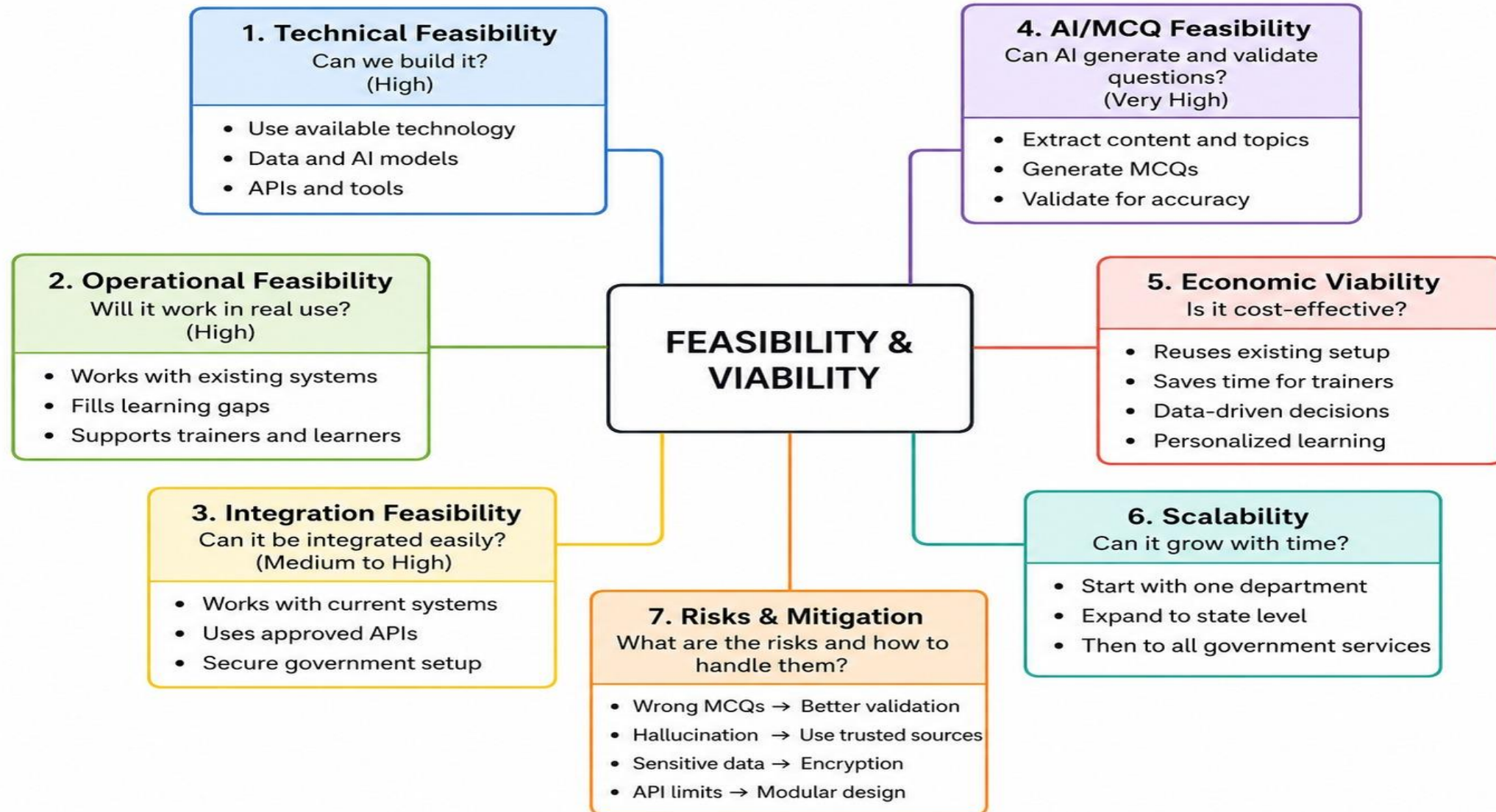


Methodology / Workflow

Technologies Used

Frontend	React.js, HTML5, CSS
Backend	Node.js , Express.js
Database	PostgreSQL
LLM/Ai	OpenAI
RAG	Langchain.js, Chroma.DB, Embeddings
Document Prasing	PDF-Parse
Integration	IGOT Karmayogi
Depolyment	Vercel





1. Enhanced Performance and Skill Development

AI-driven solutions rooted in source data can recognize gaps in competencies and offer individualized learning to bridge these gaps, thus improving the quality of skill development and assessment.

2. Personalized and Sustainable Learning

The personalized and competency-based learning combined with the continuous improvement of skills and competencies after initial development will ensure sustainability.

3. Data-driven and Future-Ready Workforce

The combination of identifying the gaps in competencies based on data and addressing these shortcomings will result in a future-ready workforce.

IMPACTS

1. Skilled and Future-Ready Workforce

The development of workers' competencies and skills through continuous upskilling and reskilling programs will make them more confident and prepared to meet future challenges.

2. Enhanced Productivity and Efficiency

Increasing the level of performances while saving time and effort by providing targeted training will enhance organizational productivity and efficiency.

3. Accessible and Scalable Digital Learning

The use of an innovative approach to enable personalized and easily accessible learning and assessment is scalable and can be used in various settings to support different learning areas and participants.

BENEFITS

Features / Aspects	Our Solution (Nexus)	Other References
Automated quiz Generation based on uploaded material	✓	✗
Adaptive Re-assessment	✓	✗
Competency Mapping	✓	✓
iGOT course Integration	✓	✓
Chat Bot	✓	✗
Personalized Trainings	✓	✓
Progress Tracking	✓	✗

❑ REFERENCES

- *“Development of a Document Processing Module for an AI Assistant Based on LLM and RAG,” IEEE SCM, 2026.*
- *“A Multimodal Retrieval-Augmented Generation System for Intelligent Document Processing and Study Assistance,” IEEE AIDE, 2026.*
- L. Ren, K. Lee and L. May, *“A Systematic Review Exploring AI's Role in Self-Regulated Learning Within Education Contexts,” IEEE Access, 2025.*
- iGOT Karmayogi: <https://www.igotkarmayogi.gov.in>
- Karmayogi Competency Module- Capacity building commission : <https://cbc.gov.in/karmayogi-competency-model-kcm>